

Teratology

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- **Learning Outcomes**
- When you have completed this section, you should be able to:
- discuss the frequency and some causes of early spontaneous abortion
- discuss some types of birth defects and major categories of their causes
- describe some syndromes that result from chromosomal nondisjunction
- explain what teratogens are and describe some of their effects.

- **Spontaneous Abortion**
- Most miscarriages are *early spontaneous abortions*
- occurring within 3 weeks of fertilization
- Such abortions are easily mistaken for a late & unusually heavy menstrual period
- 25% to 30% of blastocysts fail to implant
- 42% of implanted blastocysts die by the end of the 2nd week
- 16% of those that make it through 2 weeks are seriously abnormal & abort within the next week
- 61% of early spontaneous abortions were due to chromosomal abnormalities

- fetuses that spontaneously abort later in development show a significantly higher incidence of neural tube defects, cleft lip & cleft palate compared to newborn or fetuses of induced abortions
- Spontaneous abortion may in fact be a natural mechanism for preventing the development of nonviable fetuses or the birth of severely deformed infants

- **Birth Defects(congenital anomaly)**
- is the abnormal structure or position of an organ at birth, resulting from a defect in prenatal development
- study of birth defects is called **teratology**
- Birth defects are the single most common cause of infant mortality
- Not all of them are noticeable at birth; some are detected months to years later

- **Mutagens and Genetic Anomalies**
- Genetic anomalies are the most common known cause of birth defects,
- accounting for an estimated one-third of all cases & 85% of those with an identifiable cause
- cause of genetic defects:
- **1. mutations**
- or changes in DNA structure
- Among other disorders, mutations cause achondroplastic dwarfism , microcephaly (abnormal smallness of the head), stillbirth & childhood cancer
- Mutations can occur through-
- a. errors in DNA replication during the cell cycle
- b. under the influence of environmental agents called **mutagens**, including some chemicals, viruses, & radiation

- May genetic mutations are expressed by morphological abnormalities
- These mutations can be of dominant or recessive genes of either the autosomes or sex chromosomes(e.g. haemophilia, muscular fibrosis , cystic fibrosis)
- If gene produces its effect, whether present either upon one or upon both chromosomes of a pair of chromosome, it is called **dominant gene**
- If it produces its effect only when it is present on both chromosomes, it is called **recessive gene**

- diseases are inherited through both dominant and recessive genes called dominant inheritance and recessive inheritance, respectively
- inheritance occurring through genes of autosomes is called **autosomal inheritance**,
- inheritance occurring through genes of sex chromosomes is called **sex-linked inheritance**
- inheritance which occurs due to interaction of gene and environmental factors, viz., infectious agents, drugs, or ionizing radiation is called **multifactorial inheritance**

- **Autosomal Inheritance**
- **Autosomal Dominant Inheritance**
- When one parent is affected (i.e., have an abnormal dominant allele) then 50% children are affected
- examples of disorders caused by autosomal inheritance are:
 - Marfan's syndrome
 - Achondroplasia
 - Brachydactyly
 - Neurofibromatosis
- unaffected members of family do not transmit the trait further

- **Autosomal Recessive Inheritance**

- disease is transmitted by a couple, both of whom are carriers of an abnormal gene, but they themselves are not affected
- only siblings are affected but parents are normal
- risk of children affected is 25% in carrier couple
- E.g. sickle cell anemia

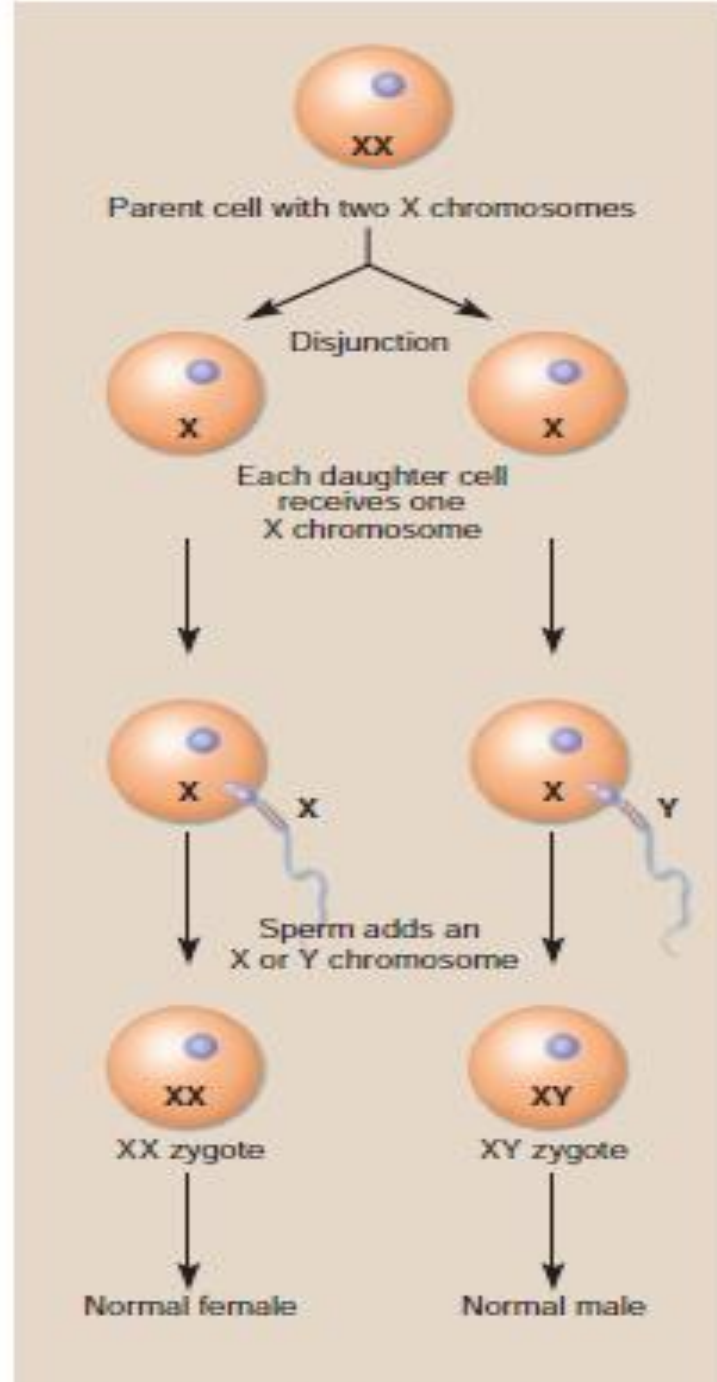
- **2. Chromosomal abnormalities**
- Usually classified as **structural or numerical errors**
- These arise during cell division especially meiosis
- Numerical errors of chromosomes result in -**aneuploidy**(total number of chromosomes other than normal 46)
- **aneuploidy** an abnormal number of chromosomes in the zygote

- **Sex-linked Inheritance**
- genes on sex chromosomes are called **sex linked**
- on X chromosome are X linked,
- on Y chromosome are Y linked

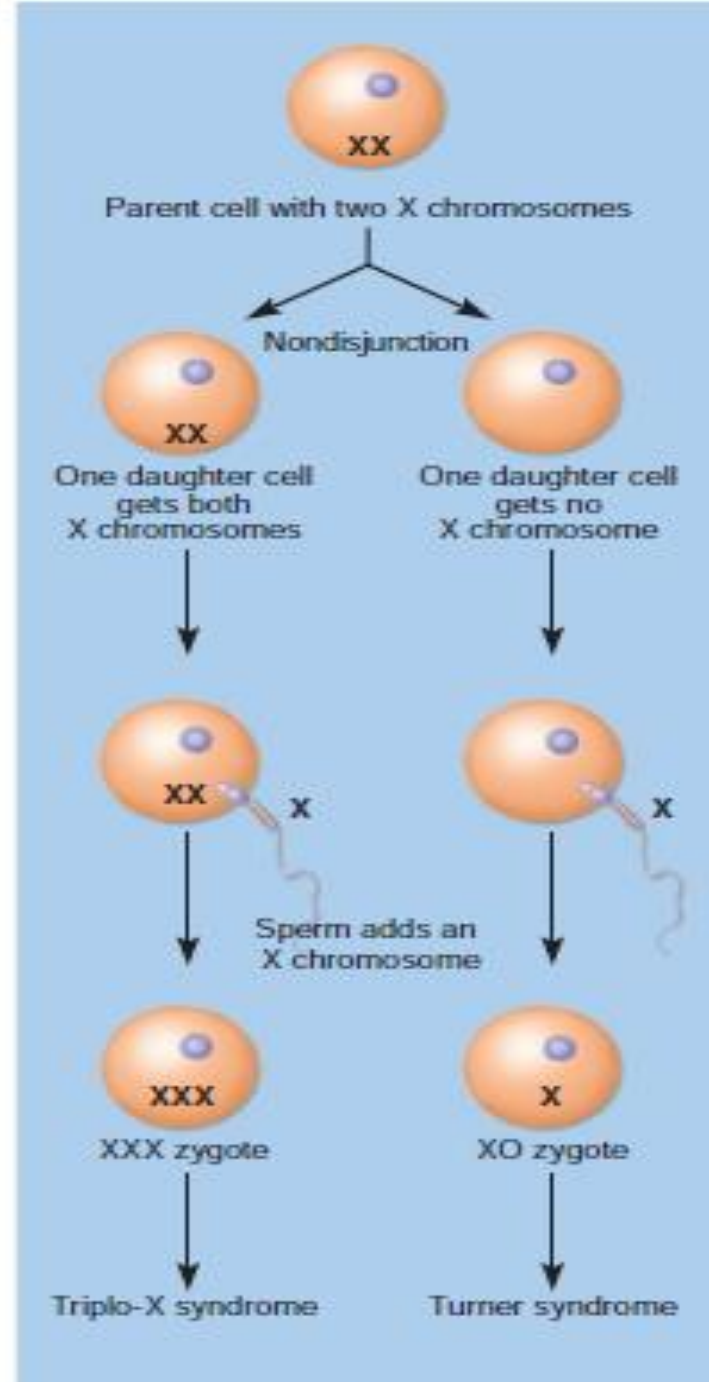
- **Abnormal numbers**
- **Polyploidy**
- Condition in which chromosomal number is a higher multiple than 2 of the haploid number (23) of chromosomes.
- In most cases, polyploid embryos abort spontaneous early in pregnancy
- High percentages of spontaneous aborted fetuses show major chromosomal abnormalities
- Polyploidy especially triploidy is likely to be caused either by the fertilization of an egg by more than one sperm or the lack of separation of a polar body during meiosis

- Aneuploidy results from **nondisjunction**
- **nondisjunction** -a failure of one of the 23 pairs of chromosomes to separate during meiosis 1 or 2, so that both members of the pair go to the same daughter cell
- **Monosomy and Trisomy**
- Monosomy-(the lack of one member of a chromosome pair)
- Trisomy- (a triplet instead of a normal chromosomal pair)
- They typically as a result of non disjunction during meiosis
- When this happens one gamete shows monosomy & the other shows trisomy of the same chromosome
- In most cases embryos of monosomy of autosomes or sex chromosomes are not viable

- E.g. suppose nondisjunction resulted in an egg with 24 chromosomes instead of the normal 23.
- If this egg were fertilized by a normal sperm, zygote would have 47 chromosomes instead of the usual 46
- In nondisjunction, an egg cell may receive both X chromosomes
- If it is fertilized by an X-bearing sperm, the result is an XXX zygote & a suite of defects called the **triplo-X syndrome**
- Triplo-X females are sometimes infertile & may have mild intellectual impairments
- syndrome often goes undetected until puberty, when failure to develop secondary sex characteristics may prompt genetic testing



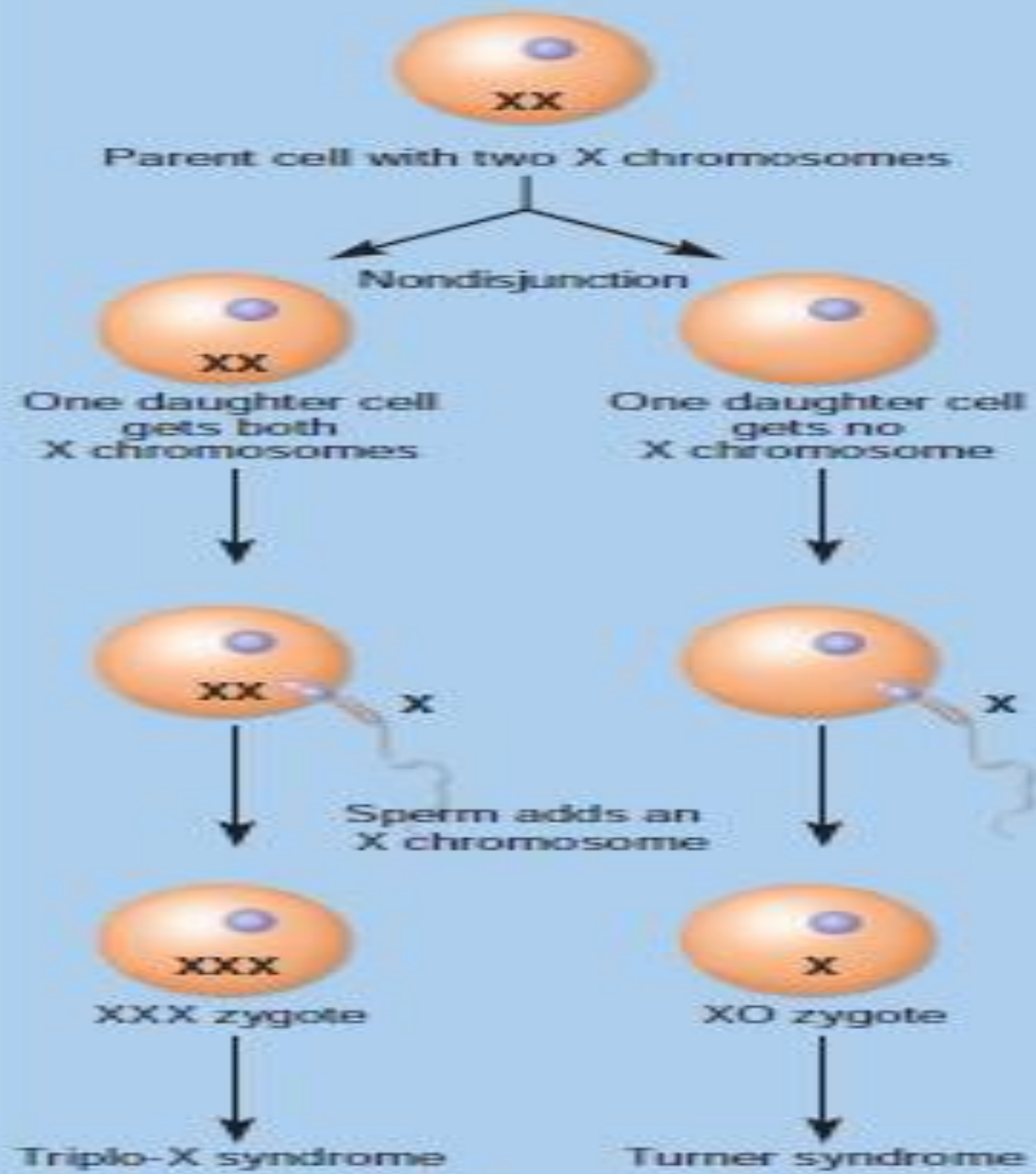
(a) Normal disjunction of X chromosomes



(b) Nondisjunction of X chromosomes

- If an XX egg is fertilized by a Y-bearing sperm, the result is an XXY combination, causing **Klinefelter syndrome**
- People with Klinefelter syndrome are sterile males, usually of average intelligence, but with undeveloped testes, sparse body hair, unusually long arms & legs, and enlarged breasts (*gynecomastia*)

- other possible outcome of X chromosome nondisjunction is that an egg cell may receive no X chromosome (both X chromosomes are discarded in the first polar body)
- If fertilized by a Y-bearing sperm, such an egg dies for lack of the indispensable genes on the X chromosome
- If it is fertilized by an X-bearing sperm, however, the result is a female with **Turner syndrome**, with an XO combination (*O* represents the absence of one sex chromosome)
- Only 3% of fetuses with Turner syndrome survive to birth. Girls who survive show no serious impairments as children, but tend to have a webbed neck and widely spaced nipples
- At puberty, the secondary sex characteristics fail to develop
- ovaries are nearly absent, the girl remains sterile & she usually has a short stature

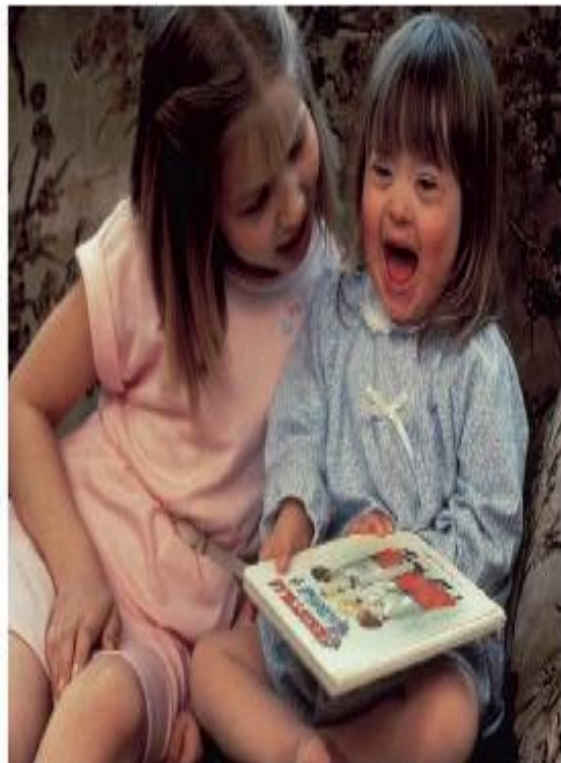


(b) Nondisjunction of X chromosomes

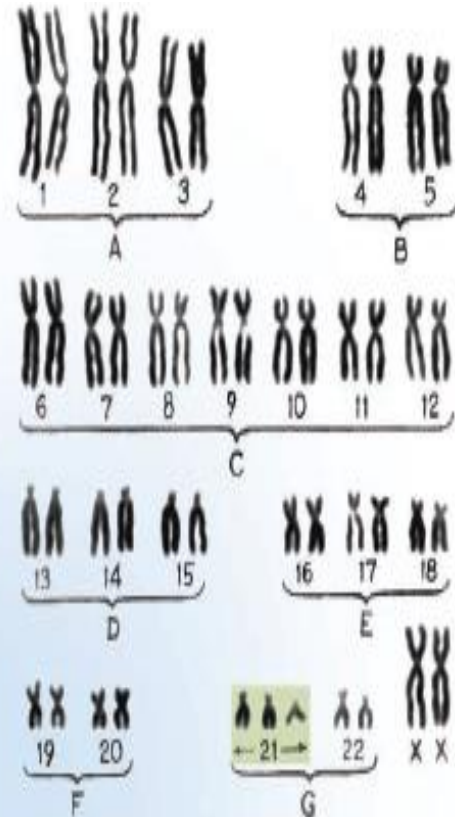


- other 22 pairs of chromosomes (the *autosomes*) are also subject to nondisjunction
- Nondisjunction of chromosomes 13 & 18 results in **Patau syndrome (trisomy-13) & Edward syndrome (trisomy-18)**, respectively
- Affected individuals have 3 copies of the respective chromosome
- Nearly all foetuses with these trisomies die before birth
- Live-born infants with these syndromes are severely deformed & fewer than 5% survive for one year

- Most common autosomal anomaly is **Down syndrome (trisomy-21)**
- Its signs include:
- retarded physical development; short stature; a relatively flat face with a flat nasal bridge; low-set ears; *epicanthal folds* at the median corners of the eyes; an enlarged, protruding tongue; stubby finger & a short broad hand with only 1 palmar crease



(a)



(b)

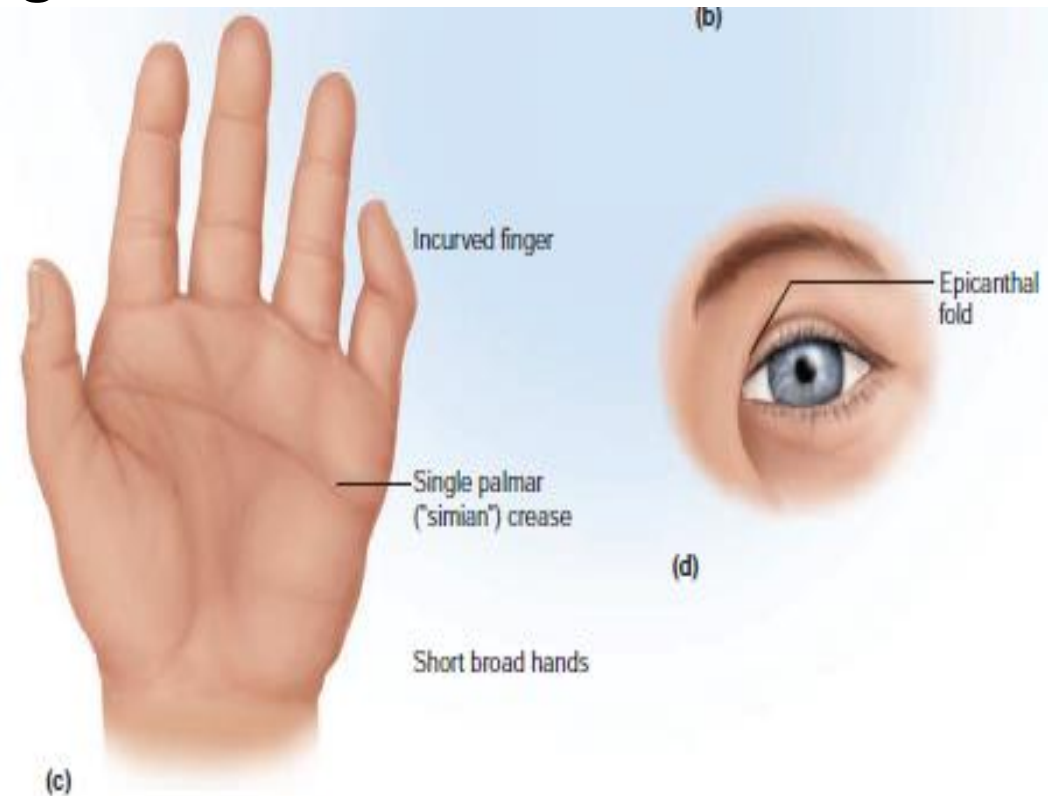


Figure 4.13 Down Syndrome. (a) A child with Down syndrome (right) and her sister. (b) The karyotype (chart of the chromosomes arranged in pairs) seen in Down syndrome, showing the trisomy of chromosome 21. (c) Characteristics of the hand seen in Down syndrome. (d) The epicanthal fold over the medial commissure (canthus) of the left eye.

• What was the sex of the person from whom the karyotype in part (b) was obtained?

- People with Down syndrome tend to have outgoing, affectionate personalities
- Mental retardation is common & sometimes severe, but is not inevitable.
- Down syndrome occurs in about 1 out of 700 to 800 live births
- Typical causes of death include immune deficiency and abnormalities of the heart or kidneys
- After the age of 40, many develop early-onset Alzheimer disease, linked to a gene on chromosome 21

- **Abnormal Chromosomal structure**
- Various abnormalities of chromosomes structures can give rise to malformations in development
- Some chromosomal abnormalities result from chromosome breakage induced by environmental factors such as radiation and chemical teratogens
- This type of structural error is usually unique to a given individual and is not transmitted to the succeeding generation
- Other type of structural abnormalities of chromosomes are generated during meiosis and if present in the germ cells can be inherited
- Common types of errors in chromosome structure are reciprocal, translocation & deletions

- E.g. congenital malformation resulting from deletion in the short arm of chromosome 5 is the cri-du-chat syndrome
- Infants with this syndrome are :
- severely mentally retarded, have microcephaly , make a cry that sound like a mewling of a cat

- **3. Teratogens**
- agents that cause anatomical deformities in the foetus
- They fall into 3 major classes: drugs & other chemicals, radiation & infectious diseases
- The effect of a teratogen depends on the genetic susceptibility of the embryo, dosage of the teratogen & time of exposure
- Teratogen exposure during the first 2 weeks usually does not cause birth defects, but may cause spontaneous abortion

- Teratogens can exert destructive effects at any stage of development, but the period of greatest vulnerability is weeks 3 through 8
- Major structural anomalies are unlikely to occur after the 8th of pregnancy because by this point, most organs would have become well established
- Different organs have different critical periods
- E.g. limb abnormalities are most likely to result from teratogen exposure at 24 to 36 days & brain abnormalities from exposure at 3 to 16 weeks
- A. most notorious teratogenic drug is thalidomide, a sedative
- Caused many born with unformed arms or legs & often with defects of the ears, heart & intestines



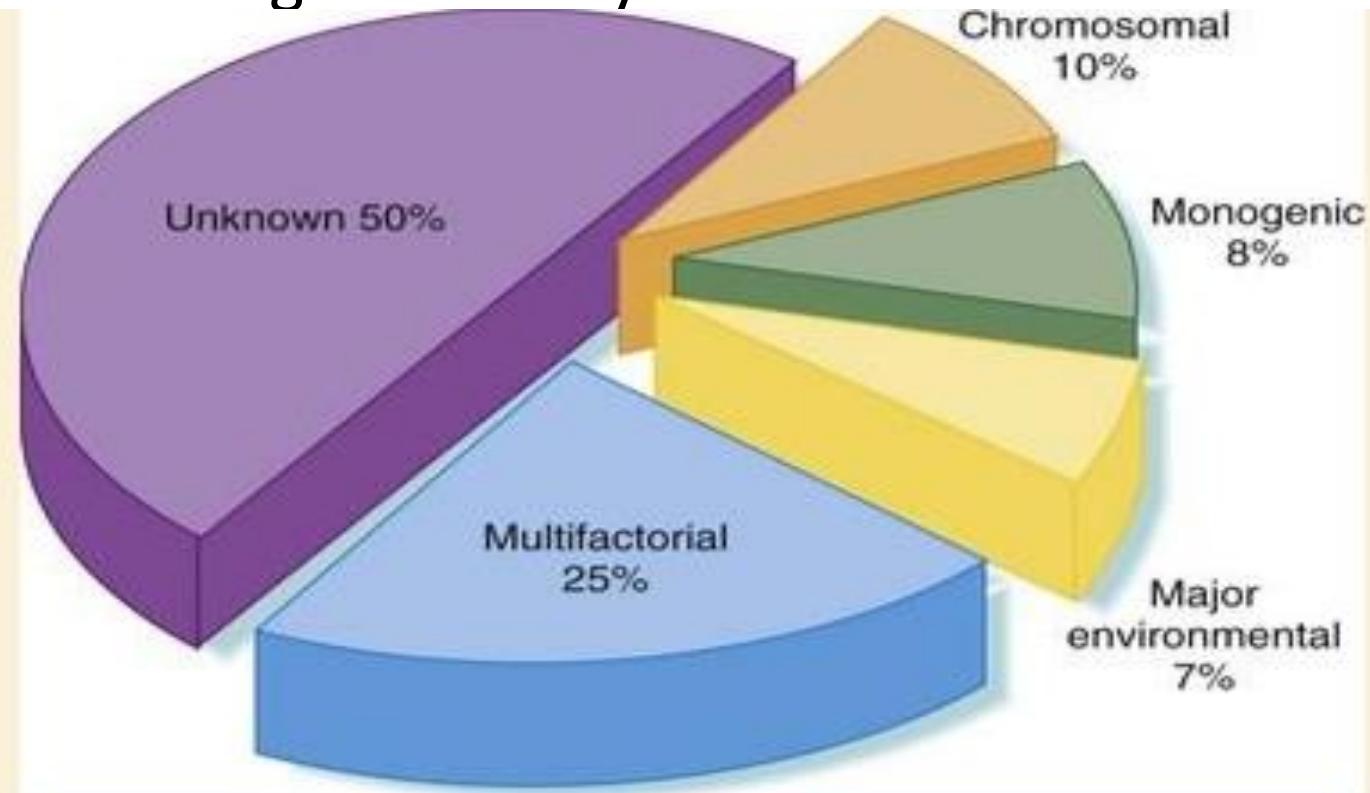
Figure 4.14 The Teratogenic Effects of Thalidomide. The infant in

- B. Alcohol abuse during pregnancy can cause **foetal alcohol syndrome (FAS)**
- characterized by a small head, malformed facial features, cardiac and central nervous system defects, stunted growth, and behavioural symptoms such as hyperactivity, nervousness, & a poor attention span
- C. Cigarette smoking also contributes to foetal and infant mortality, ectopic pregnancy, anencephaly (failure of the cerebrum to develop), cleft lip & palate & cardiac abnormalities

- **D. Infectious diseases**
- several microorganisms can cross the placenta and cause serious congenital anomalies, stillbirth, or neonatal death
- Common viral infections of the fetus and newborn include:
- herpes simplex, rubella, cytomegalovirus, and human immunodeficiency virus (HIV)
- Congenital bacterial infections include gonorrhoea and syphilis
- *Toxoplasma*, a protozoan contracted from meat, unpasteurized milk & house cats, is another common cause of foetal deformity

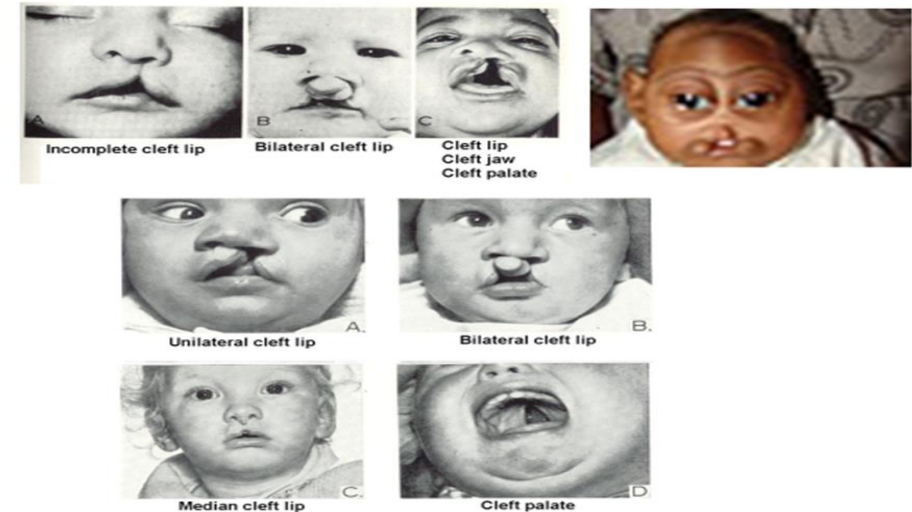
- immature immune system causes the foetus to be is vulnerable to devastating effects such as blindness, hydrocephalus, cerebral palsy, seizures & profound physical and mental retardation

- 50% of human congenital malformations remains unknown
- 18% of malformations can be attributed to genetic causes (chromosomal defects or mutations based on mendelian genetics)
- 7% of malformations are caused by environmental factors such as physical or chemical teratogens
- Of all malformations 25% are multifactorial e.g. caused by environmental factors acting on genetic susceptibility



8.7 Major causes of congenital malformations.

- **Some Disorders of Human Development**
- **Anencephaly**
- Lack of a forebrain due to failure of the cranial roof to form, leaving the forebrain exposed
- exposed tissue dies & the foetus is born (or stillborn) with only a brainstem
- Live-born anencephalic infants are very short-lived
- **Cleft lip and palate**
- Failure of the right and left sides of the lip or palate to fuse medially, resulting in abnormal facial appearance, defective speech & difficulty suckling



- **Clubfoot (talipes)**
- Deformity of the foot involving an ankle bone, the talus
- sole of the foot is commonly turned medially & as a child grows, he or she may walk on the ankles rather than on the soles
- **Cri du chat**
- A congenital anomaly due to deletion of a portion of chromosome 5
- Infants with cri du chat have microcephaly, congenital heart disease, profound mental retardation, and a weak catlike cry

- **Hydrocephalus**

- Abnormal accumulation of cerebrospinal fluid in the brain
- When it occurs in the foetus, the cranial bones separate, the head becomes abnormally large, and the face looks disproportionately small
- May reduce the cerebrum to a thin shell of nervous tissue
- Fatal for about half of patients but can be treated by inserting a shunt that drains fluid from the brain to a vein in the neck.

- **Meromelia**

- Partial absence of limbs such as the lack of some digits, a hand, or a forearm.
Amelia -Complete absence of a limb
- Phocomelia-condition in which the hand and feet seem to arise almost directly from the shoulder and hips



Fig. 8.2 Phocomelia in all four limbs.
This fetus had not been exposed to thalidomide.

- In what sense can spontaneous abortion be considered a protective mechanism?
- 13. What is the difference between mutation and nondisjunction?
- 14. Name and describe two birth defects resulting from nondisjunction of autosomes and two from nondisjunction of sex chromosomes.
- 15. Name three distinctly different classes of teratogens and give one example from each class.

- 1. A teratologist is studying the cytology of a foetus that aborted spontaneously at 12 weeks. She concludes that the foetus was triploid. What do you think this term means? How many chromosomes do you think she found in each of the foetus's cells? To produce this state, what normal process of human development apparently failed?
- A young woman finds out she is about 4 weeks pregnant. She tells her doctor that she drank heavily at a party 3 weeks earlier, and she is worried about the possible effects of this on her baby. If you were the doctor, would you tell her that there is serious cause for concern? Why or why not?